

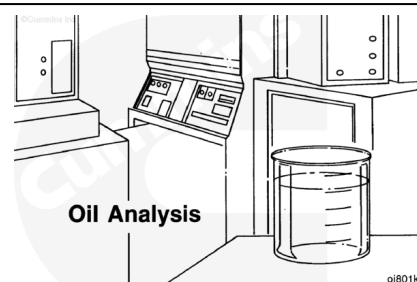
Trainings ▾

Inspect

Lubricating Oil

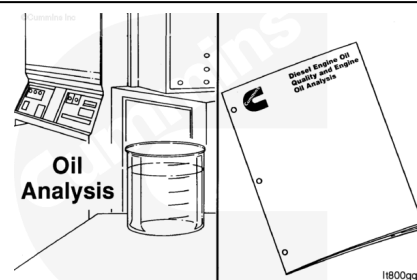
An analysis of used oil can help diagnose internal engine damage and determine if it was caused by one of the following:

- Intake air filter malfunction
- Coolant leaks
- Oil diluted with fuel
- Metal particles causing wear.



For additional oil analysis information, reference Cummins Engine Oil Recommendations, Bulletin 3810340 (</qs3/pubsys2/xml/en/bulletin/3810430.html>).

Note : Do not disassemble an engine for repair based **only** on the results of an oil analysis. The oil filter must also be inspected. If the oil filter shows evidence of internal damage, find the source of the problem and repair the damage. Refer to the appropriate procedure(s) based on the following oil filter inspection.

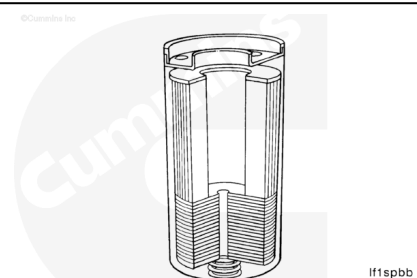


Lubricating Oil Filter

⚠ WARNING ⚠

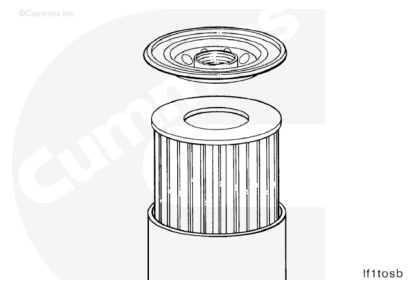
Carefully cut open the upper section of the combination filter. The filter element spring is under compression and can cause personal injury.

Use the tube cutter, Part Number 3376579, to open the upper section of the combination filter.



Inspect the filter element for evidence of moisture or metal particles.

Metal	Probable Source
Copper	Bearings and bushings
Chromium	Piston rings
Iron	Cylinder liners
Lead	Bearing overlay material
Aluminum	Piston wear or scuffing



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



The instructions below apply when visible metal particles are found in the oil filter after cutting it open.

Replace the following:

- Lubricating oil cooler element. Refer to Procedure 007-083 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-083.html>)
- Lubricating oil filter bypass. Refer to Procedure 007-014 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-014.html>)
- Lubricating oil thermostat. Refer to Procedure 007-039 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-039.html>)

Clean the following:

- Verify parts are kept clean if removed.
- **Only** use contact cleaner to clean parts.
 - Lubricating oil cooler housing. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-003-tr.html>)
 - Lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>)

Remove and inspect the following:

- Lubricating oil pump for reuse. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
- Number 1 and number 4 connecting rod bearings. Refer to Procedure 001-014 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-014-tr.html)
 - If damage is found on number 1 and number 4 connecting rod bearings, install new bearings on connecting rod journals.
 - If damage is **not** found on number 1 and number 4 connecting rod bearings, replace **only** number 1 and number 4 connecting rod bearings.
- Inspect crankshaft journals for reuse. Reference Crankshaft Reuse Guidelines, Bulletin 5411180
(/qs3/pubsys2/xml/en/bulletin/5411180.html).
- Roll new main bearings in on all seven main journals and new thrust bearings on number 4 main journal. Refer to Procedure 001-007 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-007-tr.html)

- Install lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Install lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html)
- Prime and fill the engine with new lubricating oil.
- Operate the engine to normal operating temperature and check for leaks and proper operation.

